KENYA MEDICAL TRAINING COLLEGE
FACULTY OF CLINICAL SCIENCES
DEPARTMENT OF CLINICAL MEDICINE

FINAL QUALIFYING EXAMINATION
FOR
DIPLOMA IN CLINICAL MEDICINE & SURGERY

PAPER: SURGERY

DATE: 15th February, 2023 **TIME: 3 HOURS (9:00 A.M – 12 NOON)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. All multiple choice questions **MUST** be answered in **CAPITAL LETTERS**
4. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
5. Ensure that all examination answer scripts are handed in at the end of the examination
6. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

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SURGERY

SECTION 1: MULTIPLE CHOICE QUESTIONS - (4O MARKS)

1. The most common hernia in females is;
   a) Spigelian hernia
   b) Obturator hernia
   c) Direct inguinal hernia
   d) Femoral hernia
2. The following relates to acute appendicitis;
   a) Viral infection
   b) Primary Clostridial infection
   c) Acute gastroenteritis
   d) Obstruction of the appendiceal lumen
3. The most helpful diagnostic imaging for intestinal obstruction is;
   a) Supine/erect plain abdominal x-ray
   b) CT scan of the abdomen
   c) Barium meal/swallow
   d) Abdominal ultrasound
4. Airway and cervical spine mobilization; choose the **false** statement.
   a) Maintain cervical spine immobilization in all unconscious or symptomatic patients with cervical collar.
   b) Inspect the chest movements and any signs of respiratory distress.
   c) Inspect the mouth for any debris and sweep through
   d) Perform chin left/jaw thrust to prevent the tongue from obstructing the airway.
5. Which of the following enhances wound healing?
   a) Malnutrition
   b) Diabetes mellitus
   c) Vascular sufficiency
   d) Steroids
6. Which of the following statements is true in regards to abscess?
   a) Incision and drainage is mandatory.
   b) Pus usually extends below the fascia layer.
   c) High doses of antibiotics will dry up the abscess.
   d) After incision and drainage, stitch the incision wound immediately.
7. The following modalities can be used in the main management of hydrocephalus **except**;
   a) Ventriculostomy
   b) End to end anastomosis
   c) Ventriculoperitoneal shunt
   d) Ventriculoatrial shunt
8. The following is a primary brain injury;
   a) Epidural haematoma
   b) Subdural haematoma
   c) Brain concussion/contusion
   d) Subarachnoid haematoma

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9. Abdominal pain on the right iliac fossa;
   a) May suggest a urinary tract infection.
   b) An associated mass could originate from the prostate gland.
   c) Is an indication of liver failure
   d) Acute appendicitis is a differential.
10. Which of the following is correctly matched with its gauge?
   a) Blue - Gauge 18.
   b) Yellow - Gauge 24.
   c) Orange - Gauge 22.
   d) Green – Gauge 20.
11. In management of trauma patients;
   a) The 'golden hour' interventions prevent secondary injuries and death.
   b) Multiple injuries refers to injuries to one system.
   c) Tension pneumothorax is progressive accumulation of blood in the pleura.
   d) Fracture ribs is managed by Plaster of Paris.
12. Thermal burns include;
   a) Scalds
   b) Petrol burns
   c) Naked flame burns
   d) All of the above.
13. Which of the following is the true 3rd degree burns?
   a) Spontaneous healing in one week.
   b) Blister formation.
   c) Leathery skin appearance.
   d) Severe pain on presentation.
14. In surgical history taking;
   a) General parameters include general appearance, nutritional status.
   b) History of presenting illness will be written in the patient's own words.
   c) PMSH include 'patient is a known diabetic on treatment'
   d) The specific age need not be indicated, especially in adults.
15. Choose the correct statement regarding Glasgow Coma Scale (GCS).
   a) The minimum score is 3.
   b) The maximum score is 17
   c) Eye opening to pain is graded as 3
   d) Uttering inappropriate words is graded as 4.
16. In fractures;
   a) Comminuted fractures have less than one fragment.
   b) Greenstick fractures are common in the elderly population.
   c) Impacted fractures are always displaced.
   d) In Varus deformity, the distal segment is angled towards the midline of he body.
17. Which nerve injury is associated with fracture mid-shaft humerus?
   a) Ulna nerve.
   b) Median nerve.
   c) Radial nerve.

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d) Axillary nerve.

18. Fracture complications;
    a) Avascular necrosis is common on the femoral condyles.
    b) Includes myositis ossificans
    c) Fat embolism produced cerebral vascular accident.
    d) Mal-union occurs when there is no sign of healing after >3-6 months.
19. A flail chest;
    a) Is characterized by paradoxical breathing.
    b) Is also called pigeon chest.
    c) Must be urgently stabilized and fixed with Kirschner Wires.
    d) Is always associated with haemopneumothorax.
20. Closed fracture reduction;
    a) Cannot be achieved by traction.
    b) May fail if soft tissues are interposed.
    c) Is not applicable in intra-articular fractures.
    d) Is not necessary in un-displaced fractures.
21. Internal fixation may be achieved with the following **except**;
    a) Interlocking nail
    b) Plates and screws
    c) Continuous balance fraction.
    d) Tension band wiring.
22. In management of an open fracture;
    a) The pins of an external fixator are introduced through the fracture site.
    b) Early skin closure is necessary to prevent infection.
    c) Non-viable muscles appear pink and bloody.
    d) Small bony fragments with no attachment should be removed.
23. Cardiac tamponade is characterized by;
    a) Low blood pressure.
    b) Elevated intracranial pressure
    c) Paradoxical pulse
    d) Collapsed neck veins.
24. A Surgical airway may include/involve;
    a) Tracheostomy
    b) Nasal prongs
    c) Nasal cannulae.
    d) Chest tube thoracostomy.
25. Biliary stones;
    a) Are symptomatic in majority of patients.
    b) Pigment stones are composed of purely cholesterol from excess cholesterol production.
    c) Forms as a result of super saturation of the bile.
    d) Are surgical emergency and must be removed as soon as possible.
26. Acute cholecystitis;
    a) Always requires immediate cholecystectomy

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b) Is associated with a positive Murphy's sign.
c) Cyanosis is a consistent sign.
d) May mimic a urinary tract infection.

27. Dinnerfork deformity is a feature of;
a) Osteoarthritis
b) Fracture scaphoid
c) Rheumatoid arthritis
d) Colles fracture

28. Acute urinary retention;
a) Presents with urine volume of ≥ 1 Litre.
b) Is less painful than chronic retention.
c) A suprapubic cystotomy is useful in emergency relief.
d) Phemosis is the commonest cause especially in the elderly.

29. In spinal cord injuries;
a) Brown-sequard syndrome presents with ipsilateral loss of proprioception and motor function.
b) Anterior cord syndrome causes bilateral loss of proprioception.
c) Posterior cord syndrome causes complete loss of vibration sense.
d) Cauda equine syndrome involves injury to the cervical nerve roots.

30. In controlling bleeding from the scalp;
a) Excise all avulsed scalp flaps.
b) Apply constant pressure to the bleeding area.
c) Splint the skull fractures with external appliances.
d) Apply tourniquet.

31. Features of hypothyroidism exclude;
a) Slower heart rate.
b) Cold intolerance.
c) Constipation.
d) Palpitations.

32. Benign Prostrate Hypertrophy;
a) Is always symptomatic.
b) High levels of PSA are diagnostic.
c) Is best assessed on a trans-rectal ultrasound scan.
d) Nocturia results from an associate cancer.

33. Septic arthritis;
a) Serum C-reactive proteins are reduced.
b) Cause can be iatrogenic
c) Arthrotomy is not indicated.
d) Acute osteomyelitis is a rare complication.

34. The following injury complicates to avascular necrocis;
a) Colles fractures.
b) Fracture shaft of femur.
c) Posterior dislocation of hip joint.
d) Fracture supracondylar (humerus).

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35. Haemorrhoids;
    a) Are painful anal swellings
    b) Aging causes strengthening of the anal muscles thus are uncommon among the elderly
    c) Grade three haemorrhoids oftern bleeds but do not prolapse through the anal canal
    d) Internal haemorrhoids originate above the dentate line.
36. The following is **not** a true treatment for chronic osteomyelitis.
    a) High dose of steroid therapy.
    b) Rest of the affected limb.
    c) Saucerization.
    d) Sequestrectomy
37. Differential diagnosis of haematuria;
    a) Testicular torsion.
    b) Neoplasms of bladder, prostrate, kidney.
    c) Varicocoele
    d) Spermatocoele.
38. True of undescended testes;
    a) Trans-inguinal migration of testes occurs between age 28-40 weeks of gestation.
    b) Large for gestational age (LGA) babies and post-mature infants are the most common predisposing factors.
    c) It is possible to accurately located the exact location of the testis on palpation.
    d) Orchidopexy is the surgical treatment of choice and should be performed between 13-15 years.
39. The following is **NOT** an indication for operative treatment of fractures;
    a) Compound fractures
    b) Failed conservative management
    c) Closed stable fractures
    d) Patient with multiple injuries.
40. About compartment syndrome;
    a) Presents with reduced local temperature.
    b) Rarely presents with pain.
    c) Plaster of Paris application is required.
    d) Treated by urgent fasciotomy.

SECTION 2: SHORT ANSWER QUESTIONS - (40 MARKS)

41. Define the following terms as used in surgery. **(5 marks)**
    i. Laparotomy
    ii. Arthroplasty
    iii. Fistula
    iv. Crepitus
    v. Elective surgery

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42. List at least ten (10) things you palpate for in a patient who presents with a mass. **(5 marks)**
43. State five components of conservative management of intestinal obstruction. **(5 marks)**
44. List ten (10) differential diagnosis of neck masses. **(5 marks)**
45. State five complications of fracture pelvis **(5 marks)**
46. Outline the aspects/components of wound cleaning (surgical toileting). **(5 marks)**
47. With classification, outline five differential diagnosis of haematuria. **(5 marks)**
48. List five (5) complications of fracture femur. **(5 marks)**

SECTION 3: LONG ANSWER QUESTION - (20 MARKS)

49. a) Classify compound fractures. **(10 marks)**
    b) Describe any five (5) causes of orthopaedic disorders. **(10 marks)**

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